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Q.No

Q.No

1

1A]

The role of events in logging contract actions is an important role for in smart contracts.

with the role of events in logging contract actions, we can see all the actions taken and so which what is being happening to everyone.

1B]

Inheritance can be used to modularize smart contract code,

With inheritance the smart contract code can be modularized so that the core function is defined and if we want to change and create a function we can modularize and use inheritance.

Q.No

10

Modifiers are used to enforce security and access control in complex contracts.

modifiers are already written code function which specify how the function should be available.

we can test these modifiers in many ways like brute force or privilege escalation.

11

Before deploying token deployment in a private POA, we should test it. For that we have many comprehensive testing strategies for ensuring the success of token deployment.

We need to check the environment variables and all the API keys before any token tests.

Q.No

TE

React & web3.

we can create a react app with `npx create-react-app`.

In the react app we create a form which takes user input and sends the data in POST. Then the web3 api endpoint takes the data and then processes it into its web3 format which so it works to smart contracts.

TF

React will be listening to a web3 api webhook which sends the smart contract events data and is being updated every minute. React will read that data and display it. web3 will work with smart contracts and generate data suitable for front end.

Q.No

2

2A)

Event logging assists in ensuring that `delegate call()` doesn't result in unauthorized contract behaviour or fund mismanagement.

With Event logging, every single activity is tracked and visible to the public. And it's immutable, so the logs can't be changed also.

So then when `delegate call()` function is called we can ensure that the functionality is not broken and it works as it should. and with all audits we can also ensure security and Role based access control. With all of these in place, the logging system logs every single user's activity.

Q.No

2B/Lottery Contract.

When we create a lottery contract, we create it with all functions and then upload to the smart contract. It is immutable, so once coded and uploaded we can't do anything to fix it.

When the choice between self destruct and constructor based state resetting affects transparency, because when self destruct is pressed all the gas is transferred and the contract is set to destroy itself so as the function named it as so, but with constructor based state resetting, it can be called only once while defining an object; but as it's constructor based it depends on what is in the constructor.

Q.No

201

Peers are failing to connect in a private clique network.

First check the audit log and see what's happening.

- Check if the person is have access control?
- Check ~~pass~~ username/passwords.
- Check if the network is active.
- Check if the network ID is correct.
- Reboot the system.
- Check if the boot mode settings are correct.
- Check if the admin is available.

With these steps we can trouble shoot the issue.

Q.No

20/

React A Web 3.

When polling for contract events, few practical tradeoffs are to be applied in terms of resource consumption and frequency of updates. Compared to using the event listeners in web 3,

In web 3, the react UI will be listening for a API webhook for data from web 3 backend, But this data needs to be updated and real time data.

So for that the frequency of updates will become longer as for the time needed to collect data and compile them into the frontend.

Q.No

3

3A

Require() and assert() are distinguished in terms of their impact on state changes.

Each function call is suited for pre conditions.

I say Require() is best suited for preconditions and assert() is better for invariant consistency checks.

As said each function has its own impact on state changes. The state changes are impacted by its own function call.

Q.No

3B/

In terms of scalability and long term maintenance, A Contract via a Constructor provide more flexibility for future updates compared to more permanent nature of self destruct. In constructor case contracts ~~we can~~ its is more feasible and flexible for future updates in terms of scalability and long term maintenance. This is said as the self destruct function sends all of the data to a recipient. destructs the contract itself. But as in contract based we can just recreate the contract with the future updates as to just change the constructor variables.

Q.No

5

SA]

React.

UI delays and missed updates are a huge loss for the user.

The use of event listeners in React helps much. Also, the event listener function will be put in the component, looking for any event to be captured. When captured, it will process the data and send it to the backend. This process will be repeated until the app is no longer active.

Polling strategy on the other hand is not so good. Event listeners offer a better handling of edge cases.

Q.No

5B/

Clique PoA network.

In setting up a private Clique PoA network, you see many potential failures. They can see failures like ~~from~~ node synchronization, genesis block configuration, or peer to peer connection problems. In setting up a private Clique PoA network, many can see potential failures. They can see failures like node synchronization, genesis block configuration or peer to peer connection problems, Role base access, access modifiers, Secure username and password, Boot node configuration, ~~for~~ UEFI boot options, ~~and~~ ~~login~~ ~~configuration~~.

Q.No

Q.No

[illegible]